INCH-POUND

MIL-DTL-16377/81
18 December 2008

DETAIL SPECIFICATION SHEET

FIXTURES, LIGHTING; BALLAST, FOR ONE 17-WATT T8 FLUORESCENT LAMP

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-16377.

All dimensions in inches. For tolerances, other than specified herein, see 3.5.2 of MIL-DTL-16377.

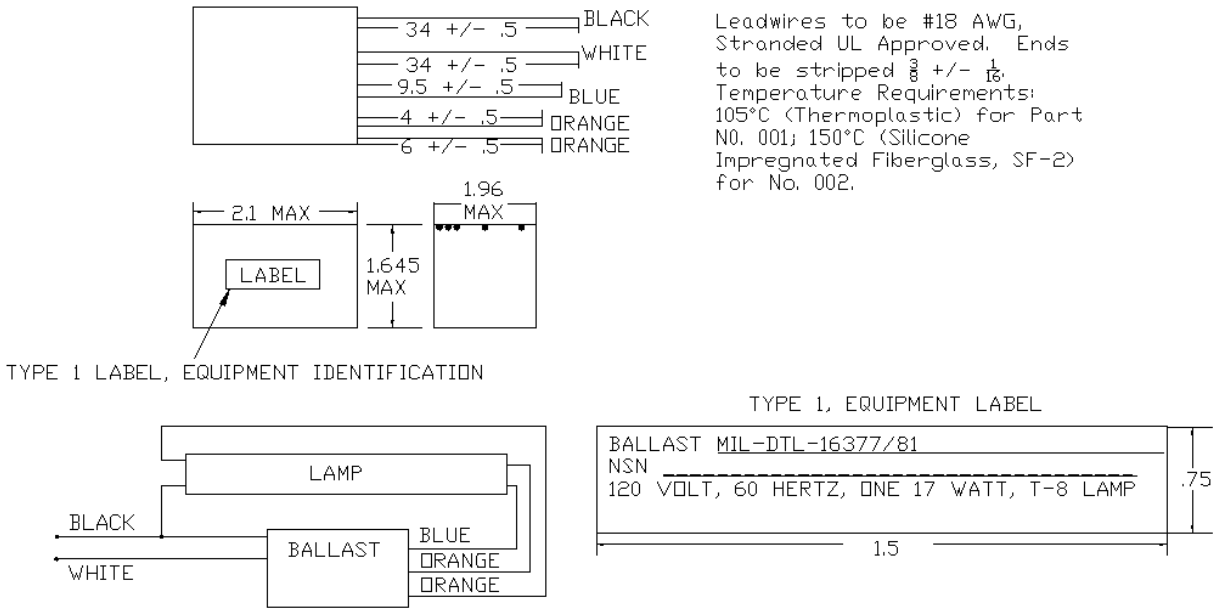


FIGURE 1. Ballast for one 17-watt T8 fluorescent lamp.

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REQUIREMENTS:

DESIGN:

Electrical rating:	17 watts, 120 volts, 60 hertz.
Power factor:	≥90 percent.
Efficiency:	>80 percent.
Circuit design:	Ballast circuit shall incorporate intergraded electronic starter for lamp per-heat starter circuit.
Dimension and configurations:	See figure 1.
Assembly:	The ballast shall consist of a laminated iron core and coil.
Impregnating:	Ballast circuitry shall be housed in thermoset plastic potting shell and potted with epoxy.

DESIGNATION AND MARKING:

One Type I label, equipment identification, reference MIL-DTL-16377 figure 11, as shown in figure 1.

EXAMINATIONS AND TESTS:

Examinations and tests shall be as specified in MIL-DTL-16377 and the following:

First article testing/inspection:	See table II.
Comparison inspection:	See table II.
Conformance inspection:	See MIL-DTL-16377.
Inspection of product for delivery:	
Examination	See MIL-DTL-16377.
Conformance test	See MIL-DTL-16377.
Examination for preparation of delivery	See MIL-DTL-16377.

GENERAL INFORMATION:

Table I. Standard parts.

Table II. First article tests.

TABLE I. Standard parts.

Name	Military p/n or specification	Remarks
Ballast, 17-watt T8 fluorescent lamp, standard	M16377/81-001	Thermoplastic insulation
Ballast, 17-watt T8 fluorescent lamp, high temperature	M16377/81-002	Silicone glass insulation

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TABLE II. First article tests and comparison inspection for ballast.[1/]

Test	Requirement	Test method	Remarks
Examination	3.13	4.5	---
Operation	3.13	4.8.1	---
Electromagnetic Interference	3.6.14	4.8.16	---
Electrical equipment interface testing requirements: MIL-STD-1399-300	MIL-STD-1399-300	MIL-STD-1399-300	2/, 3/
Dielectric withstanding voltage	3.6.15	4.8.2	4/
Insulation resistance	3.6.16	4.8.3	4/
Shock	3.5.12	4.8.7	4/
Vibration	3.5.13	4.8.8	4/
Dielectric withstanding voltage	3.6.15	4.8.2	4/
Insulation resistance	3.6.16	4.8.3	4/
Noise	3.5.14	4.8.9	---
Performance a. Power output b. Regulation c. Voltage drop across ballast d. Ballast efficiency e. Ballast power factor	5/	6/	7/

NOTES:

1/ Requirements and test methods use MIL-DTL-16377.

2/ If less, then 1KVA exception not applicable for input waveform THD.

3/ Solid state components shall be subject to 250 positive spikes and 250 negative spikes (500 total spikes), one spike per minute, using testing method identified in MIL-STD-1399-300.

4/ Tests shall be performed in the order specified.

5/ The performance of the ballast shall be in accordance with the requirements listed in the applicable paragraph of ANSI C82.2.

6/ Test procedure for performance shall be as follows:

a. Performance tests shall be made in accordance with ANSI C82.2.
b. Fluorescent lamp reference ballasts shall be in accordance with ANSI C82.3.
c. Lamp shall conform with ANSI C78.2.
d. Voltage across starter terminals and across ballast, when supplying a reference lamp, shall be measured.

7/ Sample lot size for inspection shall consist of five ballast units.

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Custodians:
Army – CR4
Navy – SH
Air Force – 99

Review Activities:
Navy – AS, CG
Air Force – 03
DLA – GS2
GSA – FAS

Preparing Activity:
Navy – SH
(Project 6250-2006-004)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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